General purpose PLL

2026 July 9

Super Long Wave

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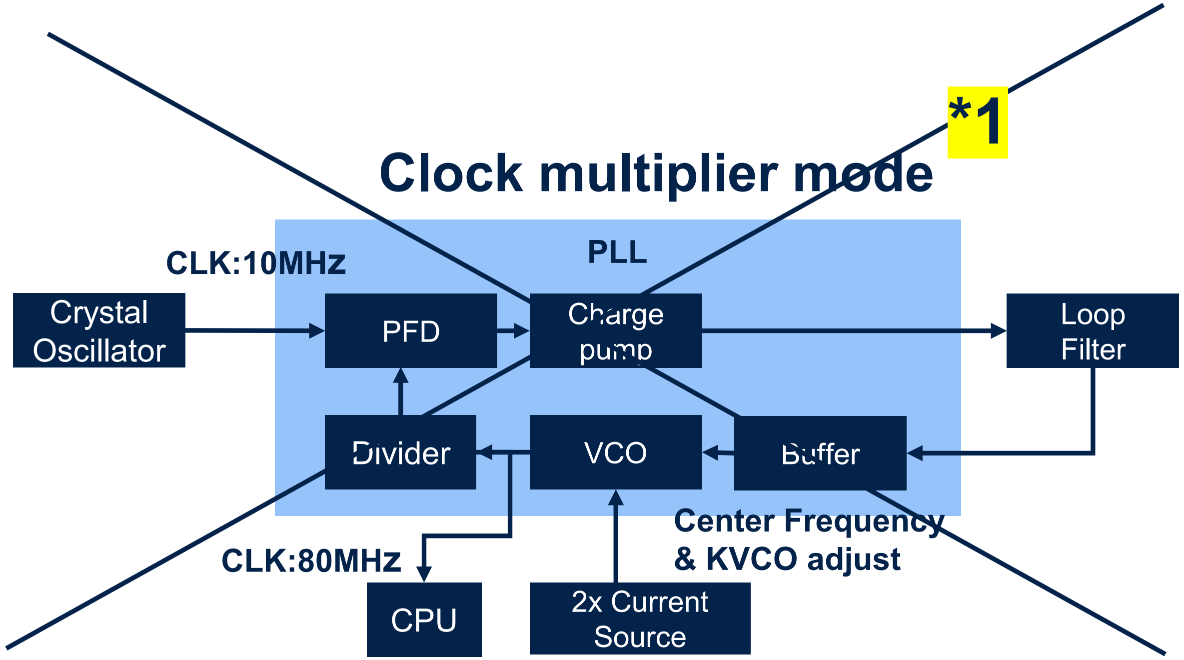
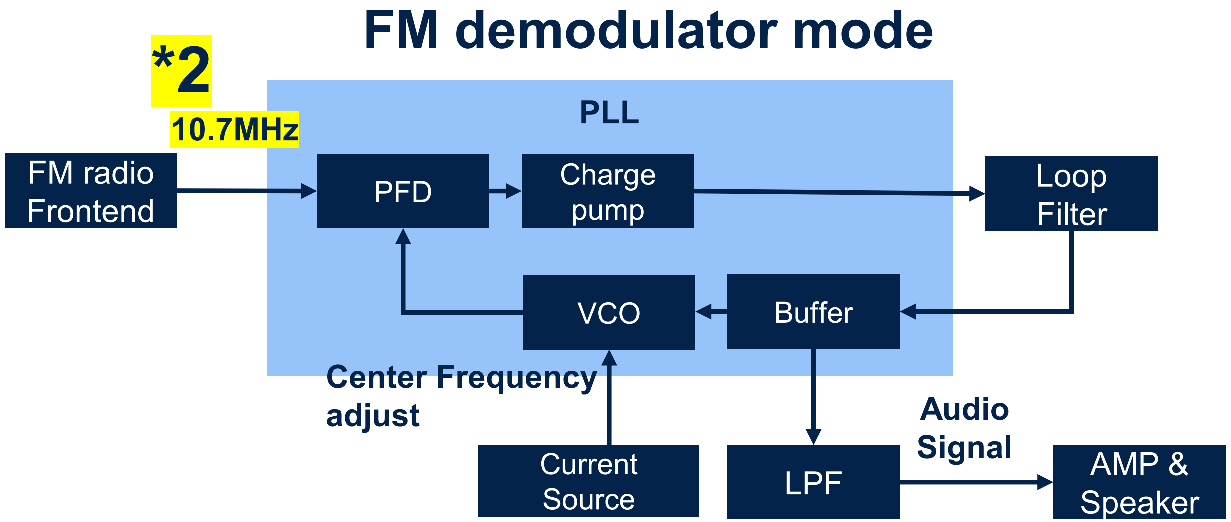
Agenda

Project information

- Project name – General purpose PLL
- Team members
  - Yutaka Kotani (team leader)
    - @Yutaka KOTANI
  - Ashutosh Chakravarty - @Teddy
  - Cheng Fei - @Buttercutter
  - Fatima - @fa_84
- Review date:2026 July 9
- Review version: git tag: v0.1
- Track C

Design objective

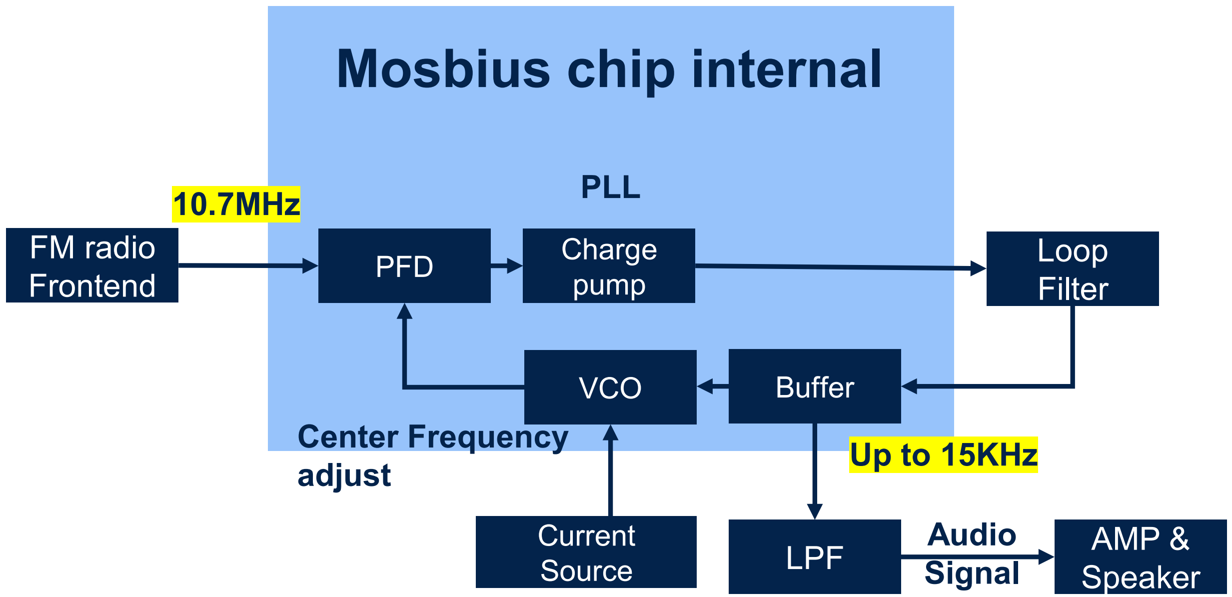




- What problems circuit solves
  - The PLL extracts the audio signal by performing FM demodulation on the 10.7 MHz intermediate frequency (IF) signal from the preceding radio front-end.
- Target specifications
  - Input FM modulated IF 10.7MHz
  - Output FM demodulated audio signal
- Intended Use
  - Create FM radio demodulator
- Project scope and spec changes
  Following our mentor's advice, we have reduced the project scope and specifications.
  1. Clock multiplier mode was deleted
  2. Input signal was changed from a direct FM signal to an intermediate frequency of 10.7 MHz

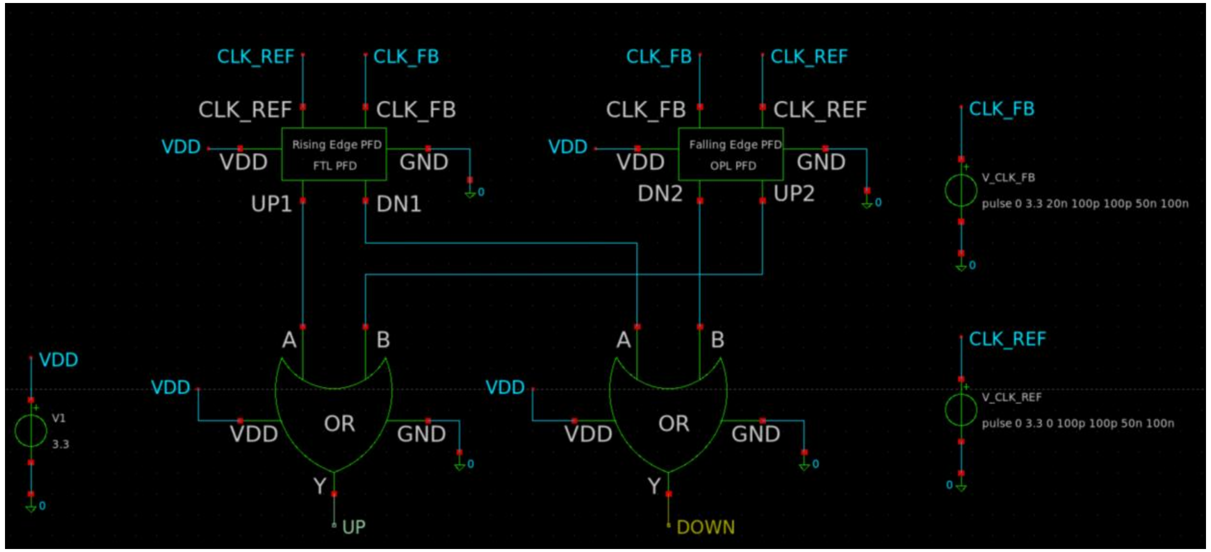
System overview

- PFD
  - Compare phase between input FM radio frontend signal and VCO output
  - Generates UP/DOWN signal to charge pump
- Charge pump
  - Converts PFD UP/DOWN signals into source/sink currents.
  - Generates the VCO control voltage through the loop filter while reducing current mismatch and reference spurs.



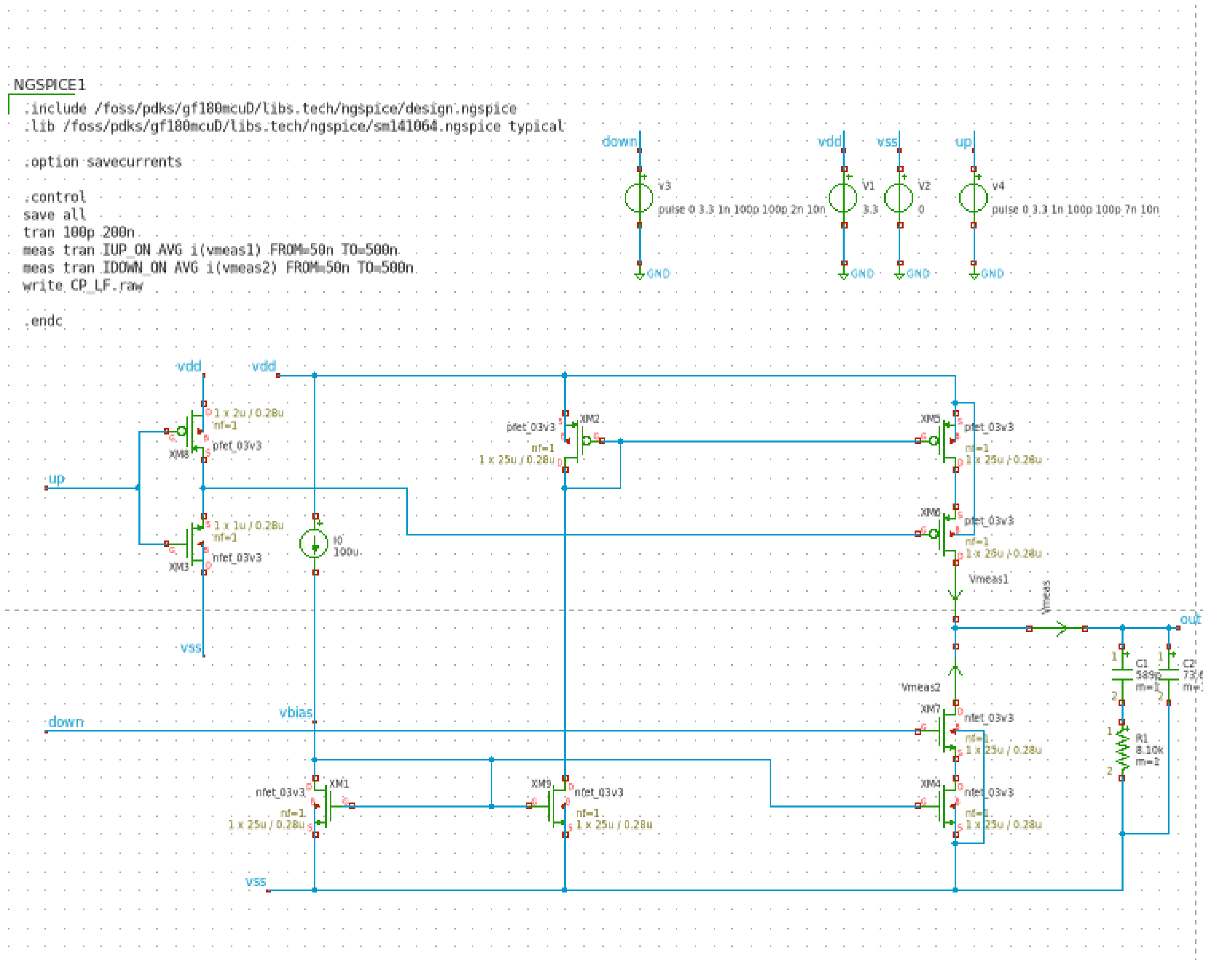
- Loop filter (external)
  - Filters the charge pump current to produce a stable VCTRL.
  - Controls loop bandwidth, stability, and lock time while reducing ripple.
- Buffer
  - To copy the voltage from the loop filter to the VCO. This ensures a reliable input
- LPF (external)
  - For removing high frequency components from buffer for smoother output
- VCO
  - Generate 0-3.3V clock signal according to input signal amplitudes
- Current source (external)
  - Provide adjustable current to set center frequency of VCO

Schematic summary - PFD



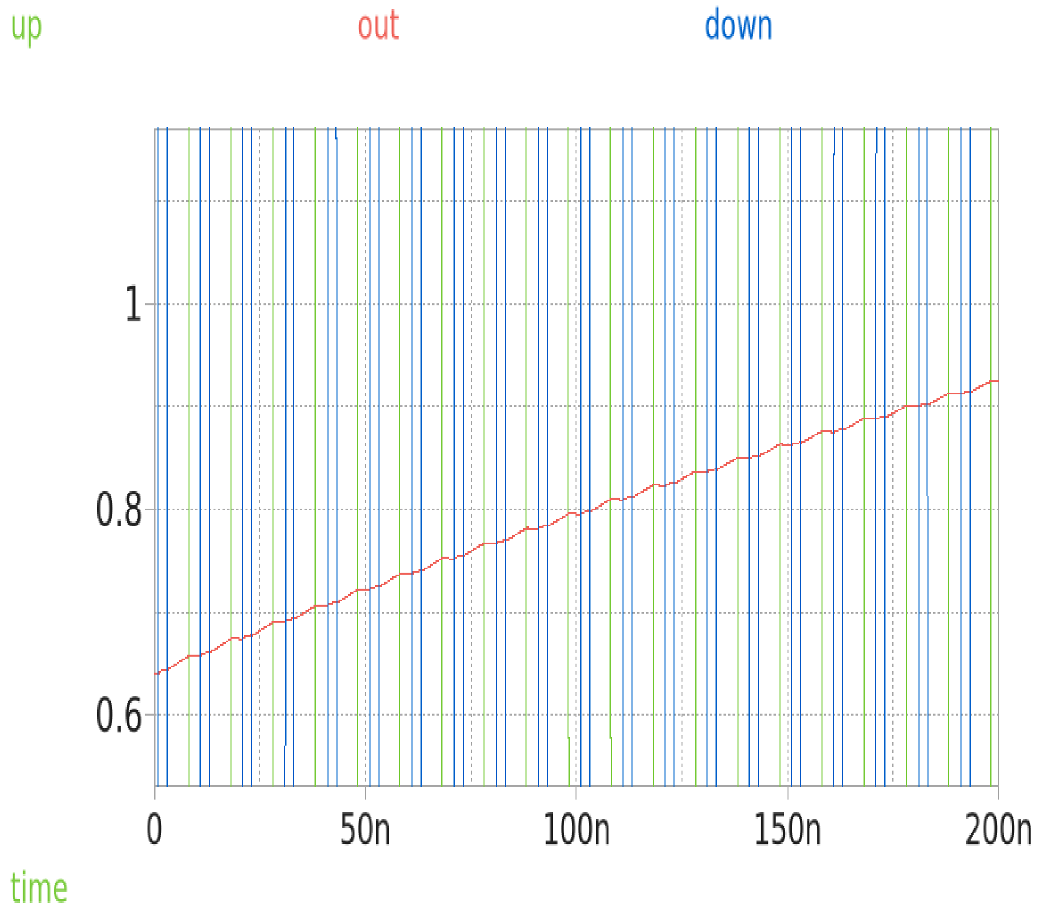
- Purpose
  - Compare phase between input FM radio frontend signal and VCO output
  - Generates UP/DOWN signal to charge pump
- Input and Output
  - Input: FM radio frontend signal and VCO output
  - Output: UP/DOWN signal
- Important design decisions
  - T.B.D
- Interface to other blocks
  - Receives rf modulated signal and generate up/down signal to charge pump

Schematic summary – Charge pump/ Loop filter



- Purpose
  - Converts PFD UP/DOWN pulses into source/sink currents that charge or discharge the loop filter, generating VCTRL for the VCO
- Input and Output
  - Input: up / down digital control signals from PFD
  - Output: charge/discharge current at “out” node, into loop filter (C1 / R1 / C2)
- Important design decisions
  - Cascaded PMOS/NMOS switch pairs (XM2/XM5/XM6 and XM7/XM4) for high output impedance
  - I_UP / I_DOWN current matching monitored directly in simulation to limit reference spurs
  - 100 µA reference current (I0) distributed via vbias to all mirror branches
- Interface to other blocks
  - Receives up/down directly from PFD; drives the external loop filter node that sets VCO control voltage

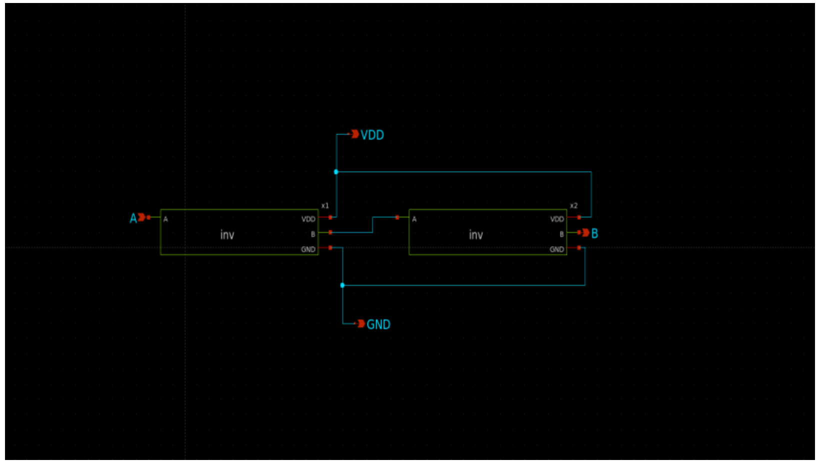
Schematic summary – Charge pump Transient simulation results

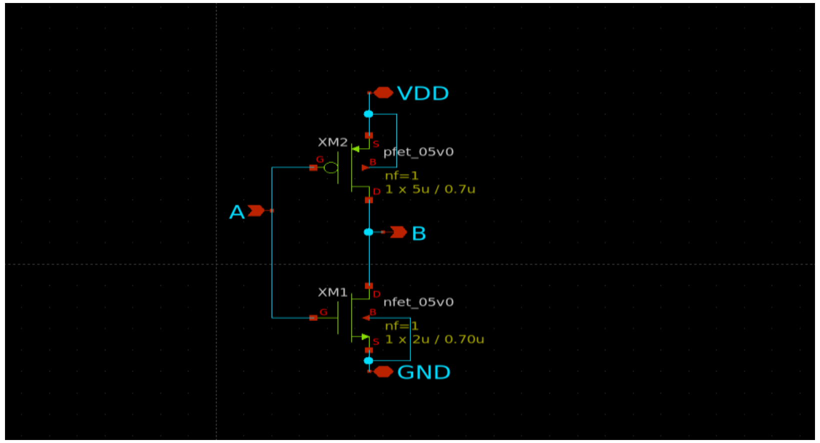


- Testbench
  - ngspice transient, GF180MCU 3.3V devices, up/down pulses (100ps edges) driving charge pump into C1/R1/C2 loop filter
  - IUP_ON_AVG and IDOWN_ON_AVG measured over 50ns–500ns to quantify current matching
- Observed behavior
  - Out node ramps smoothly from ~0.6V toward ~1V as up/down pulses accumulate charge on the loop filter
  - No visible glitching or discontinuity at each switching edge, consistent with well-matched cascode switching
- Next steps
  - Extract numerical I_UP/I_DOWN mismatch and reference-spur estimate across PVT corners

Schematic summary – Buffer

- Schematic (Buffer, Inverter respectively)





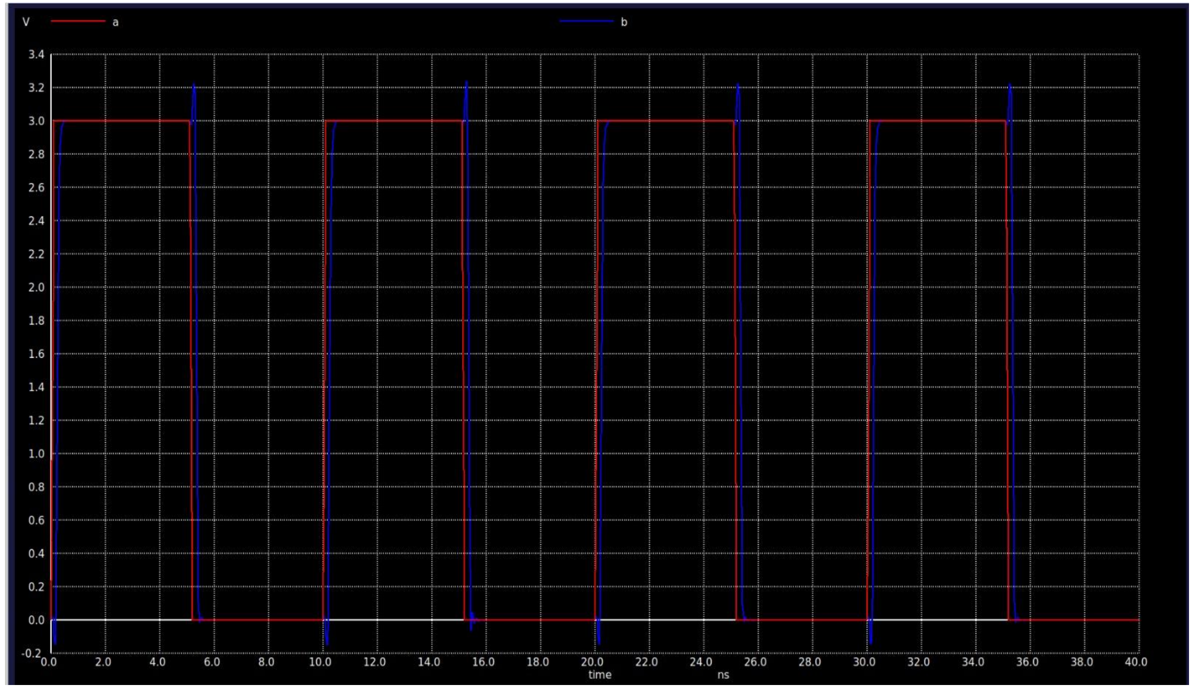
- Purpose

To copy the voltage from the loop filter to the VCO. This ensures a reliable input

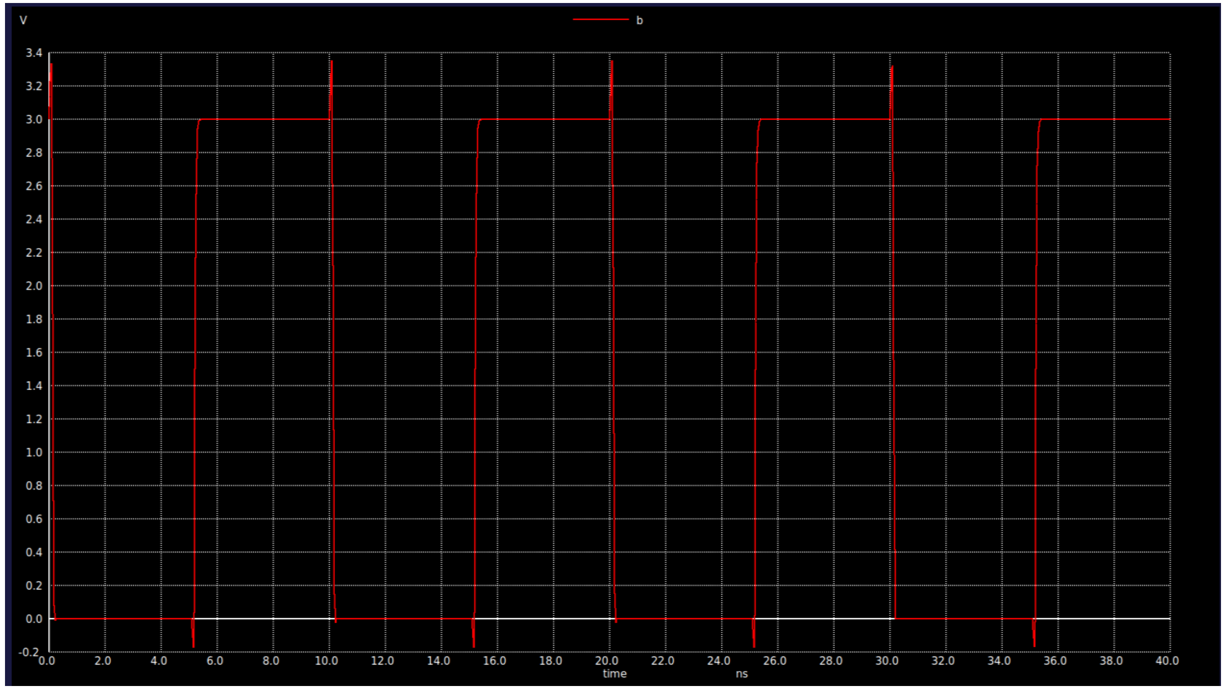
- Input and Output
  - Input 3V pulse signal (for simulation)
  - Output 3V pulse signal identical to input (simulation result)
  - 3V VDD input

Simulations of Buffer

- Output of the Buffer

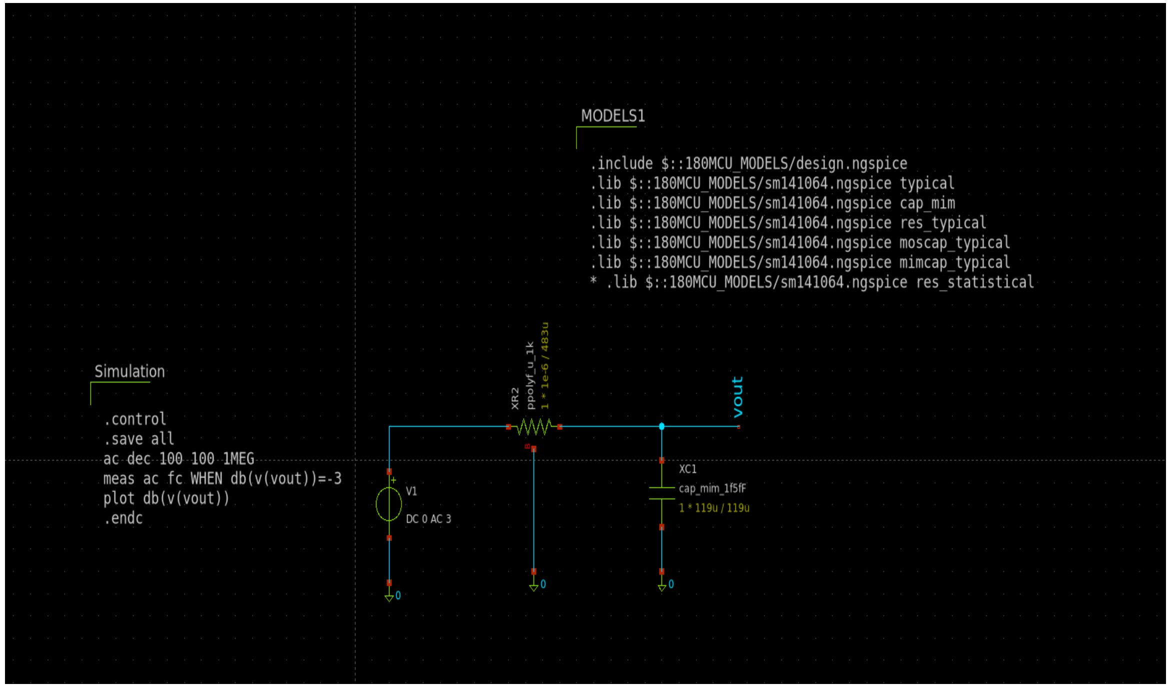


- Output Of Inverter



Schematic summary – LPF

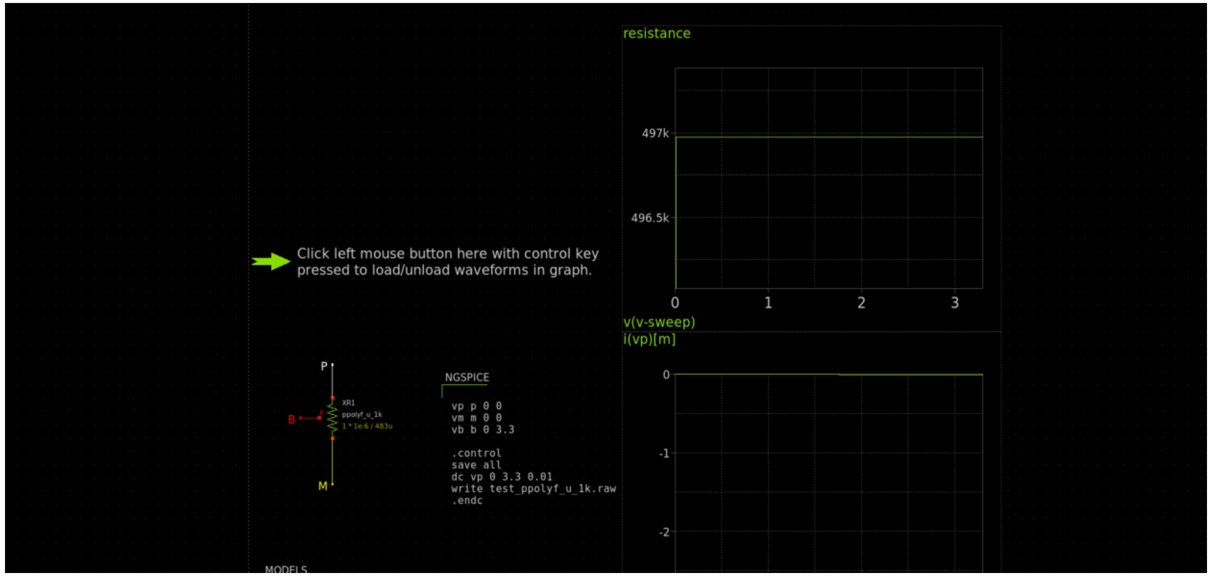
- Schematic



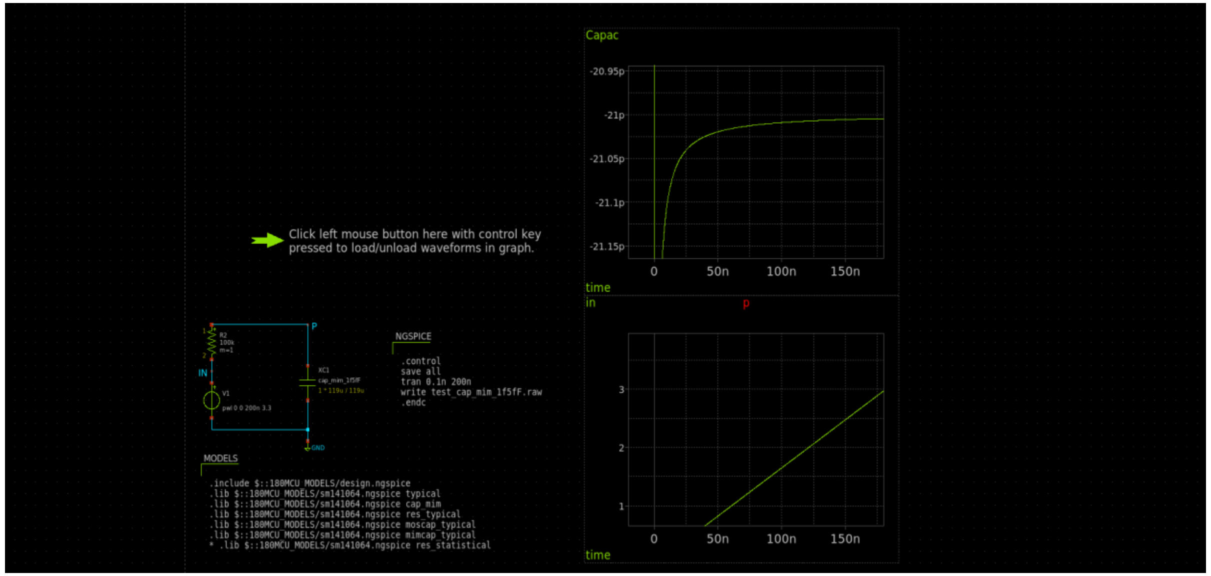
- Purpose
  For removing high frequency components from buffer for smoother output
- Input and Output
  - Buffered signal containing high frequency and noise
  - Clean filtered signal
- Important design decisions: Selection of R and C values of reasonable size for filter design
- Interface to other blocks Input from buffer

Resistor and Capacitor Value Selection

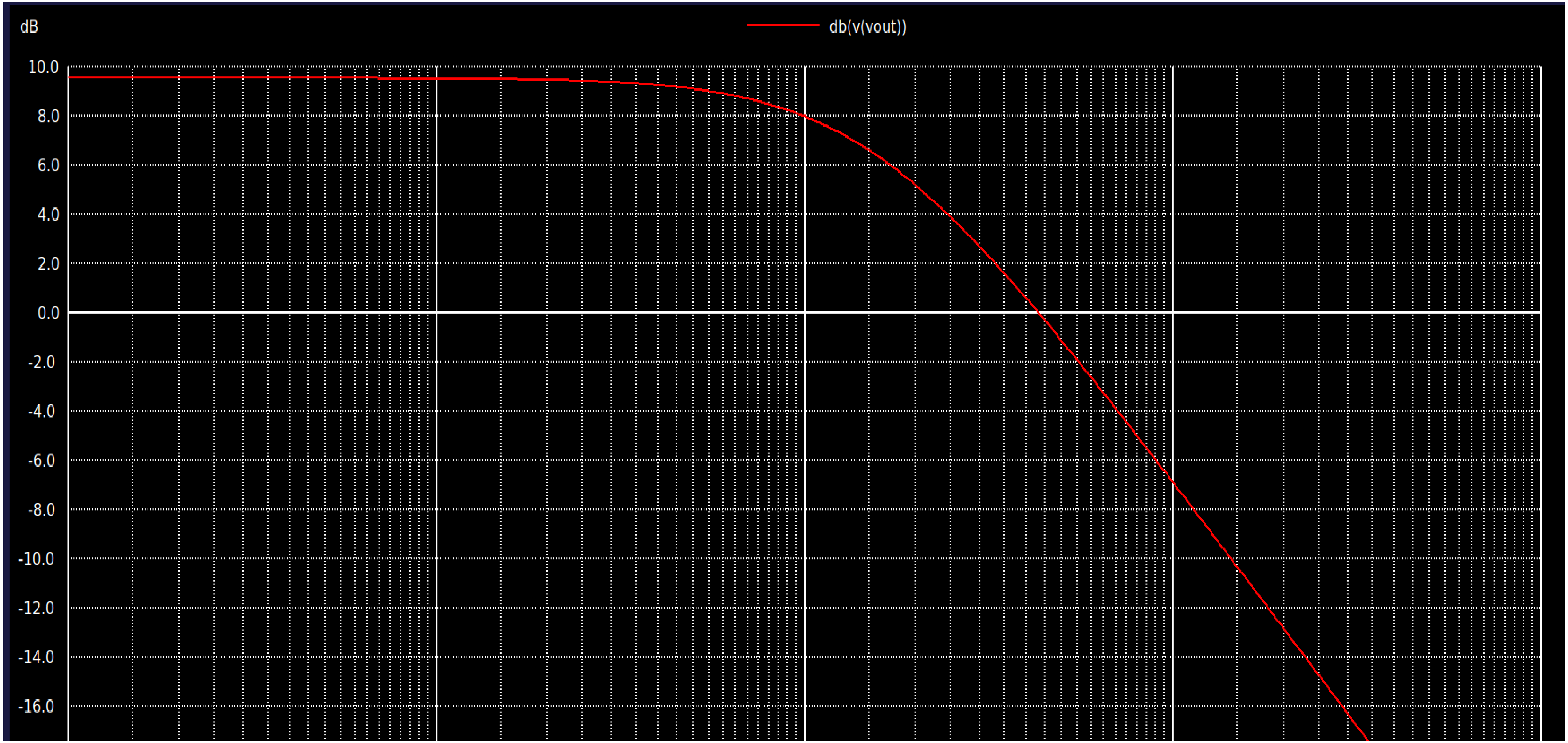
- Resistor Selection



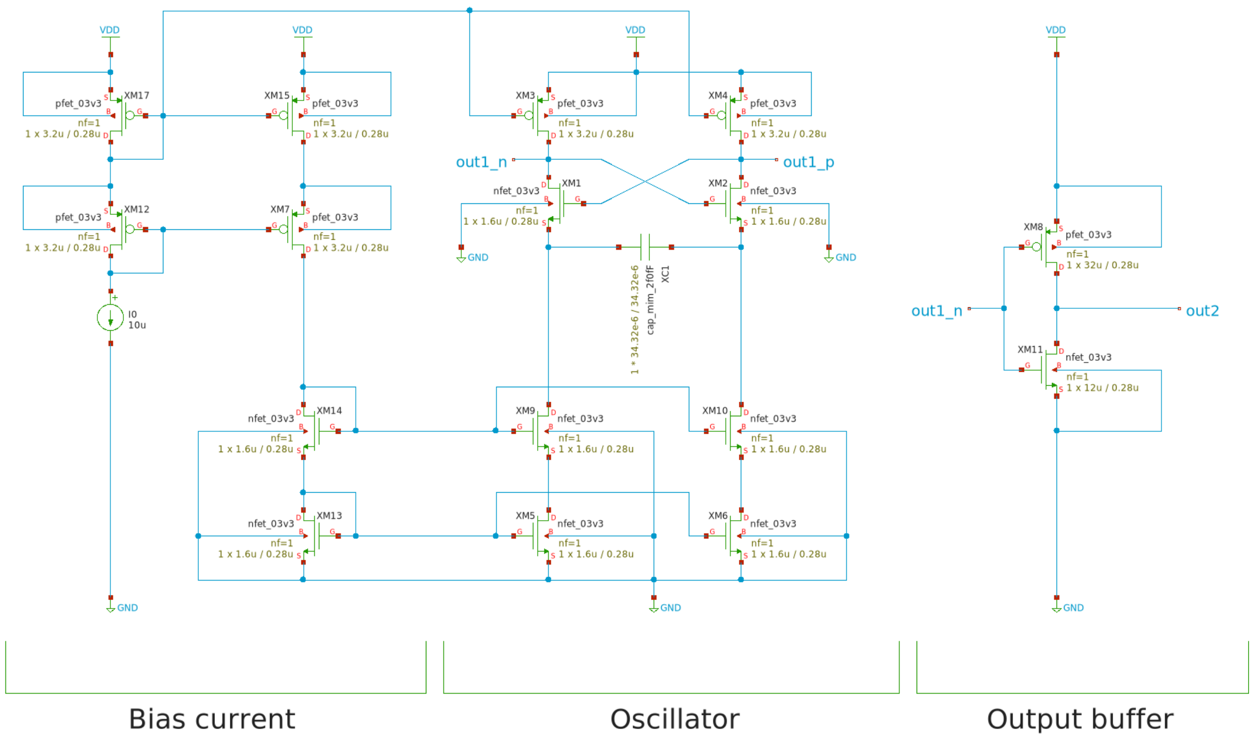
- Capacitor Selection



Simulation of LPF



Schematic summary – VCO



- Purpose
  - Generate 0-3.3V clock signal according to input signal amplitudes.
- Input and Output
  - Input: voltage signals that indicate output frequency
  - Output: 10.7MHz ±200Khz clock signal
- Important design decisions
  - Define suitable KVCO according to FM Radio spec
    - Assumption: FM radio audio frequency deviation:15KHz, Input signal max amplitude: 1V
    - Suitable KVCO:75KHz/V
  - Using just only one capacitor to tune output frequency for ease of turning
- Interface to other blocks
  - Received VCO control signal from loopfilter; Generate clock signals.

Schematic summary – current source

- schematic diagram T.B.D

- Purpose
  - Provide adjustable current to set center frequency of VCO
- Input and Output
  - Input: tunable resistor with turning knob
  - Output: generate adjustable current up to 200uA
- Important design decisions
  - Fine turning is necessary
- Interface to other blocks
  - Generate reference current to VCO.

Design assumption

Target Spec	Min	Typ	Max
Pin count		8	
Estimated Area		***um2	
pll_in voltage [V]	0.8	-	2.5
pll_out voltage [V]	0		vdd
FM demodulator mode	**Min**	**Typ**	**Max**
pll_in frequency [MHz]	10.6	10.7	10.8
dmod_out voltage [V]	0	-	vdd
Cen_freq_cur [uA]	10	-	110
PVT	**Min**	**Typ**	**Max**
Process	SS	TT	FF
Vdd voltage [V]	3	3.3	3.6
Temperature [degC]	20	25	30

Pin	Pin name	Notes
1	VDD	3.3V
2	GND	
3	PLL_IN	Reference clk input
4	PLL_OUT	VCO output
5	CEN_FREQ_CUR	Center frequency bias current
6	KVCO_ADJ_CUR	kvco adjust bias current
7	LF_IN	External Loop filter input
8	LF_OUT	External Loop filter output

Expected load	Min	Typ	Max
Load impedance [ohm]	48k	100k	330k
Load capacitance [pF]	10	22	33

Verification status

- See “project_status_tracker_super_long_wave.xlsx”

No	Main Component	Sub component	Owner	Status	Schematics	Functional simulation	Simulation Link	Design Checklist	Corner simulation	Monte Carlo	ERC	LVS	DRC
1	PFD		Buttercutter	On going	Done	On going	T.B.U	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
2	Charge pump		fa_84	On going	Done	Pass	CP_LF.sch	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
3	Loop filter (external)		fa_84	On going	Done	Pass	CP_LF.sch	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
4	Buffer		Teddy	On going	Done	Pass	tb_buffer.sch	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
5	LPF (external)		Teddy	On going	Done	Pass	tb_lpf.sch	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
6	VCO	CCO	Yutaka KOTANI	On going	Done	Pass	tb_vco_tran.sch	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
7		OTA	Yutaka KOTANI	On going	On going	Not yet	T.B.U	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
8	Current source (external)		Yutaka KOTANI	On going	On going	Not yet	T.B.U	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
9	Integration		Yutaka KOTANI	Not yet	Not yet	Not yet	T.B.U	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet

Design checklist

- See “design_checklist/design_checklist_0*_*****.xlsx”

design_checklist_01_pfd.xlsx
design_checklist_02_Charge_pump.xlsx
design_checklist_03_Loop_filter_external.xlsx
design_checklist_04_Buffer.xlsx
design_checklist_05_LPF_external.xlsx
design_checklist_06_vco_cco.xlsx
design_checklist_07_vco_ota.xlsx
design_checklist_08_current_source_external.xlsx
design_checklist_09_integration.xlsx

Design checklist

Main Component	PFD/Charge pump/Loop filter (external)/Buffer/VCO/Current source (external)/Integration
Sub component	
Type	Analog/Digital/Mixed
File name	dummy.sch
Version tag	v0.1
Designer	Yutaka KOTANI/Teddy/Buttercutter/fa_84
Reviewer	Yutaka KOTANI/Teddy/Buttercutter/fa_84
Review date	Jul-26

PASS	OK, Worked as expectation
ALERT	Need to confim design, Unexpected behavior observed
FAIL	Need to modify design,Critical issue for functionality
N/A	Not applicable

No	Category	Check item	Description	Expected behavior	Observed behavior	Result
1	Functionality	Requirements implemented				
2						
3		Reset behavior verified				
4		Start up verified				
5		Power sequencing considered				
6	Analog	Bias current checked				
7		Device operating regions verified				
8		Stability analyzed				
9		Noise considered				
10		Matching requirement considered				
11	Digital	Timing reviewed				
12		State machine behavior verified				
13		Clock crossing indentified				
14		Reset strategy documented				
15	Mixed Signal	Analog/digital interface reviewed				
16		Isolation strategy doumented				
17		Substrate and supply noise considered				
18	Reliability	Device voltage limits checked				
19		Current density reviewed				
20		Latch-up prevention considered				
21		ESD stratety identified				
22	Documentation	Nets clearly named				
23		Hierarchy consistent				
24		Parameters documented				
25		Symbols match implementation				

Open issues

- See "open_issue_super_long_wave.xlsx"

No	Description	Impact	Proposed solution	Owner	Status
1					
2					
3					
4					
5					
6					
7					
8					
9					

Questions for reviewers

- See "questions_for_reviewers_super_long_wave.xlsx"

No	Category	Priority	Question	Questioner	Answer	Answerer	Status
1							
2							
3							
4							
5							
6							
7							
8							
9							